

Learning Domain-Agnostic Spatial-Angular Feature for Light Field Image Super-Resolution

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Abstract

Extracting domain-specific features of Light Fields (LF) is a common practice in Light Field Image Super-Resolution (LFSR), typically using CNNs or Transformer on sub-aperture images (SAI), macro-pixel images (MacPI), and epipolar plane images (EPI). While these methods are efficient and effective, they often fall short in fulfilling the key LFSR requirement: *simultaneously preserving spatial details and maintaining angular consistency during processing*. In this paper, we introduce LF-DANet for LFSR, which aims to learn Domain-Agnostic spatial-angular features from LFs to improve the reconstruction quality. To this end, we first develop a Detail-Awareness Transformer Layer (DATL) as a general feature extraction unit for better preserving fine-grained details. Then, building upon DATLs, we propose a Domain-Agnostic Feature Extraction Block (DAFEB) to learn domain-agnostic spatial-angular features of LFs. Specifically, a domain-agnostic spatial-angular feature can be obtained in three steps. First, We employ DATLs to extract inter-view and intra-view spatial information on MacPIs and SAIs. Then, we enhance angular consistency by applying DATLs on EPIs. Since the second step can diminish the spatial context information of the feature, we finally reapply the previous DATLs to enhance the spatial details of the structure-constrained feature, ensuring the feature encapsulates both fine-grained spatial details and robust angular consistency. Our method achieves superior performance and visual quality on LF benchmarks with competitive efficiency.

Introduction

Light Field (LF) cameras can capture different directional light rays emitted from one point by adding a micro-lens array between the main lens and the camera sensors, thus enabling various visual applications just taking a single shot, such as depth estimation (Wang et al. 2022b; Wang, Lin, and Zhang 2021), 3D reconstruction (Cai et al. 2018), novel view synthesis (Kalantari, Wang, and Ramamoorthi 2016; Mildenhall et al. 2019), and refocus (Wang et al. 2018b). However, the trade-off between spatial and angular resolution reduces the spatial quality of sub-aperture images (SAI), limiting the usability of LF images. To break this dilemma, lots of efforts have been put forward to solve the Light Field image Super-Resolution (LFSR) in recent years, which aims to increase the spatial resolution of SAIs.

In the last decades, numerous deep neural networks have been proposed to promote the performance of LFSR. Since

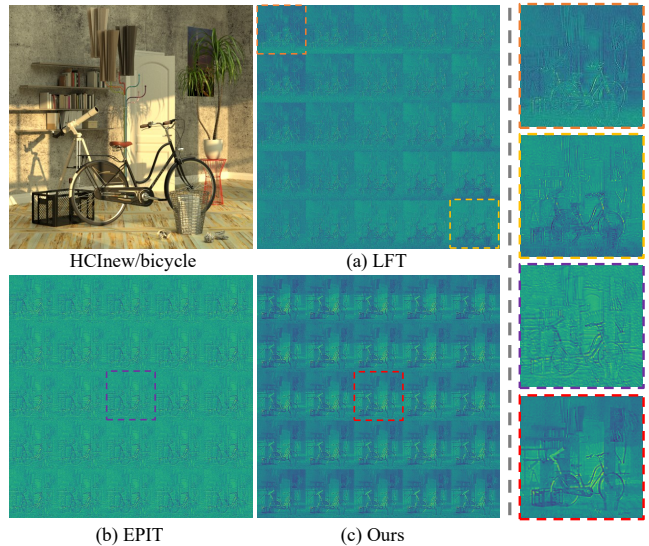


Figure 1: Illustration of the feature maps extracted from LFT (Liang et al. 2021), EPIT (Liang et al. 2023), and our method. The feature map generated by LFT presents low angular consistency (i.e., the left-up corner SAI feature focuses on the background, while the right-down corner SAI feature focuses on the foreground, e.g., bicycle and trash can), and the feature map generated by EPIT achieves high angular consistency while obtaining coarse features, both present strong domain-specific properties. In contrast, our method generates domain-agnostic features with both high angular consistency and fine-grained details.

Yeung et al. (2019) first introduced a 2D Spatial-Angular-Separable convolution to approximate the effect of 4D convolution to process 4D LFs, extracting features from different subspaces of 4D LFs (i.e., sub-aperture image (SAI), macro-pixel image (MacPI), and epipolar plane image (EPI)) has become a common practice in this field due to its efficiency and effectiveness. However, we identify a significant limitation in this approach. We observe that features extracted from a certain subspace often exhibit strong domain-specific characteristics. For instance, features extracted from SAIs and MacPIs tend to involve more spatial context information but lack strong structural constraints or

angular consistency. On the other hand, features extracted from EPIs, which are closely related to scene depth, typically capture more structural information but with less spatial details. These domain-specific tendencies reveal two key issues in current methods that employ similar strategies. First, methods that perform feature extractions in partial domains, such as LFT (Liang et al. 2021) and EPIT (Liang et al. 2023), fail to fully leverage the complementary properties of different LF subspaces, as illustrated in Fig. 1. Second, methods that extract complete domain-specific features and then fuse them, such as HLFSR (Duong et al. 2023), often result in fused features that do not adequately assemble these domain-specific characteristics as expected and can even lead to a neutralization effect, which diminishes the distinctiveness of these properties. Therefore, there is a compelling need for a novel method that can generate domain-agnostic spatial-angular features, effectively capturing both spatial details and angular consistency. Such a method is crucial for reconstructing high-quality and accurate high-resolution LF images. Additionally, we find that existing Transformer-based methods are inclined to produce blurry high-resolution (HR) images, especially in subtle textures and edge areas, missing high-frequency details. This phenomenon is attributed to the under-exploration of local dependencies and also can be explained by (Park and Kim 2022) that the self-attention mechanism exhibits low pass filter behaviors which weakens high-frequency signals.

To address these issues, we first devise a Detail-Awareness Transformers Layer (DATL) which integrates Cross-Shape Window Self-Attention (CW-SA) and a Convolutional-Gated Feed Forward Network (CG-FFN) for better extractions of local fine-grained features. Next, based on DATLs, we construct a Domain-Agnostic Feature Extraction Block (DAFEB) to learn domain-agnostic spatial-angular features from 4D LFs. Specifically, we first utilize DATLs to extract inter-view and intra-view spatial features on MacPIs and SAIs. Subsequently, we use DATLs to process horizontal and vertical EPIs to enhance the features’ angular consistency. Since the features end up with feature extraction on EPIs, the spatial context information is suppressed. Thus, we reapply the previous DATLs to enhance the spatial context information of the resulting structurally constrained features in the last step, finally producing domain-agnostic features that embody both enhanced spatial details and robust angular consistency. Building upon DAFEBs, we develop a novel network, termed LF-DANet for LFSR. In summary, the contributions of this paper are as follows: **1)** We propose a Detail-Awareness Transformer Layer to facilitate effective extractions of local fine-grained features, which improve the recovery of spatial details. **2)** Based on DATLs, we propose a Domain-Agnostic Feature Extraction Block that can extract domain-agnostic spatial-angular features from LF, which can well preserve the spatial details and angular consistency simultaneously. **3)** We develop a network termed LF-DANet based on the above designs. Our method surpasses the former state-of-the-art method LF-DET by 0.45dB in average PSNR on $\times 4$ SR with 20% less parameters and 59% reduced inference time.

Related Work

Light Field Image Super-Resolution. Light Field Image Super-Resolution (LFSR) has been attracting increasing interest in recent years due to the fast advancement of deep learning techniques. Yoon et al. (2017) first utilize CNNs to super-resolve each SAI. However, the performance of LFCNN is limited because the angular information was ignored during super-resolution. Consequently, the following works spare no effort to explore the abundant angular information to improve the performance through various strategies such as stacking SAI sequences of different directions (Wang et al. 2018a; Zhang, Lin, and Sheng 2019; Zhang, Chang, and Lin 2021), performing spatial-angular-separable convolutions (Yeung et al. 2019; Jin et al. 2020), performing 3D convolutions (Zhang, Chang, and Lin 2021; Liu et al. 2023), and using 2D convolutions on MacPI with different dilation rates (Wang et al. 2020, 2023; Duong et al. 2023). Inspired by the success of Vision Transformer (Dosovitskiy et al. 2020), some Transformer-based networks were proposed to explore long-range dependencies of LFs, which is difficult for CNNs. LFT (Liang et al. 2021) can be regarded as a Transformer version of LFSSR (Yeung et al. 2019), which employs Transformer on spatial and angular domains alternately. Wang et al. (2022a) take SAIs as sequence elements and learn long-range geometric relations on horizontal and vertical SAI sequences. Liang et al. (2023) addressed the disparity problem by learning non-local spatial-angular correlation on EPIs using Transformer. Similar to LFT (Liang et al. 2021), Cong et al. (2024) proposed a spatial-angular separable Transformer while being more efficient and robust to disparity variation.

Vision Transformer with Local Design. The original vision Transformer (Dosovitskiy et al. 2020) can establish long-range relationships between image patches. However, the local feature inside the patches is ignored when directly projected to vectors. Therefore, researchers start to inject locality into Transformers. Mainstream approaches include performing local attention instead of on the full image (e.g., Swin Transformer (Liu et al. 2021) and CSwin Transformer (Dong et al. 2021)), or hierarchical attention (e.g., TNT (Han et al. 2021), and Twins (Chu et al. 2021)). Another way is to hybridize CNNs with Transformers to capture both local and global information. Architectures such as concurrent CNN-SA token mixer (e.g., ConFormer (Peng et al. 2021), and DAT (Chen et al. 2023)), convolutional feed-forward networks (e.g., CMT (Guo et al. 2022), and LocalVit (Li et al. 2021)), and convolutional tokenizer (e.g., CVT (Wu et al. 2021), (Xiao et al. 2021)) have been demonstrated effective in enhancing the locality and improving modeling capability of Transformer architectures. However, most of the existing Transformer-based LFSR networks lack effective extraction for local features, leading to unsatisfactory reconstruction of fine-grained details.

Method

Problem Formulation. The goal of LFSR is to produce HR LF images $\mathcal{L}_{hr} \in \mathbb{R}^{U \times V \times \alpha H \times \alpha W}$ when given LR LF images $\mathcal{L}_{lr} \in \mathbb{R}^{U \times V \times H \times W}$, where α is the upscale factor.

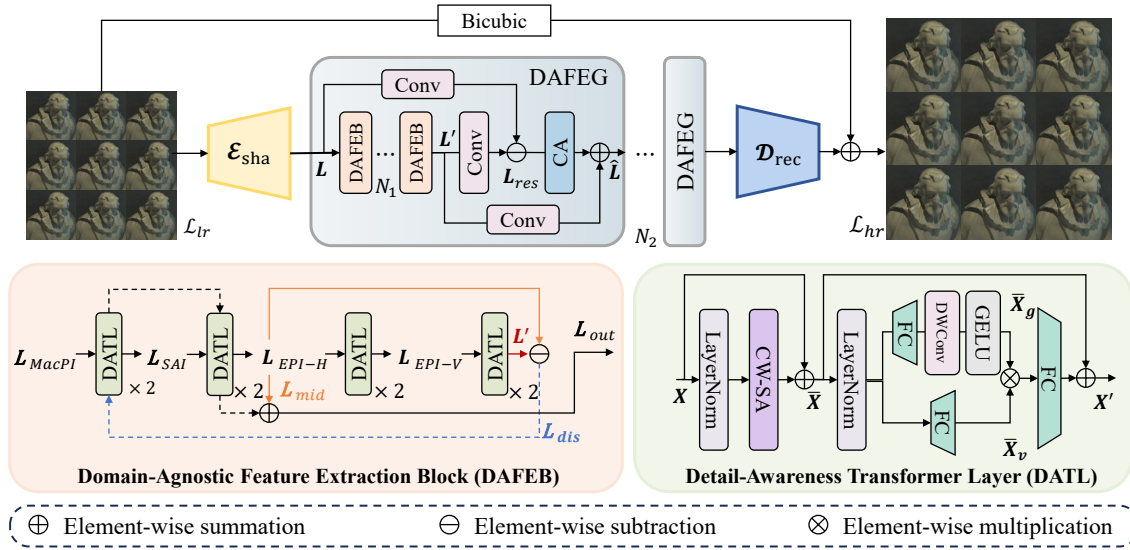


Figure 2: **Detailed architecture of LF-DANet.** The input LF $\mathcal{L}_{lr}$ is first processed by $\mathcal{E}_{sha}$ to extract shallow features. Then, N_2 cascading Domain-Agnostic Feature Extraction Groups (DAFEG) are employed, and each DAFEG consists of N_1 Domain-Agnostic Feature Extraction Blocks (DAFEBs) with a residual channel attention learning to extract domain-agnostic spatial-angular features. Finally, the high-resolution light field images $\mathcal{L}_{hr}$ are reconstructed through the reconstruction module $\mathcal{D}_{Rec}$.

Following (Wang et al. 2023), we only super-resolve the luminance component and calculate quantitative metrics on it, and the LF data is organized to an $U \times V$ SAI array as input.

Overall Architecture. The overall architecture of LF-DANet is illustrated in Fig. 2. Specifically, we first apply the shallow feature extraction module $\mathcal{E}_{sha}$ to obtain the shallow features of LF images $\mathbf{L} \in \mathbb{R}^{U \times V \times H \times W \times C}$, where C denotes the channel dimension. Next, N_2 Domain-Agnostic Feature Extraction Groups (DAFEGs) are cascaded to process the extracted shallow feature for spatial-angular learning. Finally, the obtained spatial-angular features are sent to the reconstruction module $\mathcal{D}_{Rec}$ to generate the final high-resolution light field images.

Each DAFEG is mainly composed of N_1 Domain-Agnostic Feature Extraction Blocks (DAFEBs), which are the core component of our model. N_1 DAFEBs are organized with the residual channel attention learning to enhance the information flow and preserve high-frequency details. As depicted in Fig. 2, the initial feature $\mathbf{L}$ is processed by N_1 DAFEBs to generate the enhanced feature $\mathbf{L}' \in \mathbb{R}^{U \times V \times H \times W \times C}$. Then, we extract the residual between the input and output features, which will be further improved by Channel Attention (CA) (Hu, Shen, and Sun 2018). Finally, the enhanced residual will add to the original output feature. Notably, we employ a 1×1 Conv layer in each branch to increase features' adaptivity. The above process can be formulated as

$$\begin{aligned} \mathbf{L}' &= \mathcal{F}_{DAFEB}^{N_1}(\cdots \mathcal{F}_{DAFEB}^1(\mathbf{L})), \\ \mathbf{L}_{res} &= \text{Conv}_{1 \times 1}(\mathbf{L}') - \text{Conv}_{1 \times 1}(\mathbf{L}), \\ \hat{\mathbf{L}} &= \mathcal{F}_{CA}(\mathbf{L}_{res}) + \text{Conv}_{1 \times 1}(\mathbf{L}'). \end{aligned} \quad (1)$$

Each DAFEB primarily consists of 8 Detail-Awareness Transformer Layers (DATLs). Of these, 2 DATLs are ded-

icated to extracting features from macro-pixel images, 2 DATLs focus on sub-aperture image feature extraction, and the remaining 4 DATLs are utilized for extracting features from EPI images. In the following sections, we will delve into the detailed structure of the DATL.

Detail-Awareness Transformer Layer

Despite the advancement of Transformer, we found existing Transformer-based LFSR networks reconstruct passable HR LF images when facing challenging scenarios with intricate structures, wherein this situation fine-grained local context information is crucial to recover these high-frequency details. Therefore, we propose a Detail-Awareness Transformer Layer (DATL) to intensify the locality awareness of our model. Concretely, we first utilize Cross-shaped Window Self-Attention (CW-SA) instead of full image attention to establish relationships among tokens which is helpful to alleviate the interference of distinct meaningless pixels and aggregate more useful neighbor information according to the local inductive bias of images. Specifically, after normalizing the input feature $\mathbf{X} \in \mathbb{R}^{H \times W \times C}$, we divide it into two parts $\mathbf{X}_h \in \mathbb{R}^{H \times W \times C/2}$ and $\mathbf{X}_v \in \mathbb{R}^{H \times W \times C/2}$ that $\mathbf{X}_h$ will be evenly partitioned to M horizontal stripes and $\mathbf{X}_v$ will be evenly partitioned into N vertical stripes, both with stripe width $sw = 1$. Then, we perform Multi-Head Self-Attention (MHSA) on each stripe

$$\begin{aligned} \mathbf{X}_h &= [\mathbf{X}_h^1, \cdots, \mathbf{X}_h^M], \text{ where } M = W/sw, \\ \mathbf{X}_v &= [\mathbf{X}_v^1, \cdots, \mathbf{X}_v^N], \text{ where } N = H/sw, \\ \bar{\mathbf{X}}_h^i &= \text{MHSA}(\mathbf{X}_h^i W_Q^i, \mathbf{X}_h^i W_K^i, \mathbf{X}_h^i W_V^i), \quad i \in [1, M], \\ \bar{\mathbf{X}}_v^j &= \text{MHSA}(\mathbf{X}_v^j W_Q^j, \mathbf{X}_v^j W_K^j, \mathbf{X}_v^j W_V^j), \quad j \in [1, N], \end{aligned} \quad (2)$$

where W_Q , W_K , and W_V are projection matrices. Finally, the enhanced features are concatenated together to generate

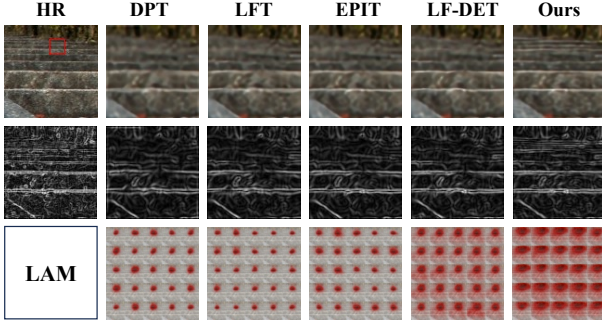


Figure 3: Comparisons of $\times 4$ SR on *Paved_Road* from EPFL dataset between LF-DANet (Ours) and existing Transformer-based methods (i.e., DPT, LFT, EPIT and LF-DET). Our method is effective in recovering local high-frequency details (shown by the extracted contours), and capturing global contextual information (indicated by the Local Attribution Maps (LAM)).

the output $\bar{\mathbf{X}} = [\bar{\mathbf{X}}_h, \bar{\mathbf{X}}_v]$.

Second, inspired by (Shi 2024), we further employ a Convolutional Gated Feed-Forward Network (CG-FFN) in the second stage. Concretely, a normalized feature after attention $\bar{\mathbf{X}}$ will be conveyed to a gating branch through FC-DWConv-GeLU to generate gating signals $\bar{\mathbf{X}}_g$, and a value branch through a FC layer to get the value $\bar{\mathbf{X}}_v$. Then, the gating signals multiply with the value to obtain enhanced feature, followed by another FC layer to produce the output. By simply adding a DWConv layer before activation in the gating branch, gating signals turn out to be adaptively fine-grained, which is beneficial to preserve import features such as edges and contours as shown in Fig. 3. In addition, by applying DATL in a high-order manner, information can be exchanged across different windows to achieve a global respective field as indicated by the Local Attribution Maps (LAM) (Gu and Dong 2021) in Fig. 3.

Domain-Agnostic Feature Extraction Block

The detailed structure of the DAFEB is illustrated in Fig. 2, given an input LF feature $\mathbf{L} \in \mathbb{R}^{U \times V \times H \times W \times C}$, the processing can be separated to three stages. First, initially enhancing the spatial details by extract intra-view spatial features and inter-view spatial features on SAI and MacPI respectively. Specifically, we first reshape the feature to MacPI representation $\mathbf{L}_{MacPI} \in \mathbb{R}^{HW \times U \times V \times C}$, which will be processed by two DATLs to integrate inter-view spatial information as

$$\mathbf{L}'_{MacPI} = \mathcal{F}_{DATL}^2(\mathcal{F}_{DATL}^1(\mathbf{L}_{MacPI})). \quad (3)$$

Then, the feature is reshaped to SAI $\mathbf{L}_{SAI} \in \mathbb{R}^{UV \times H \times W \times C}$ for intra-view spatial feature extraction as

$$\mathbf{L}'_{SAI} = \mathcal{F}_{DATL}^2(\mathcal{F}_{DATL}^1(\mathbf{L}_{SAI})). \quad (4)$$

Note that, we omit unnecessary Reshape operation for simplicity. And we will keep this in the following descriptions.

Subsequently, we begin to explore the structure information of LFs to incorporate angular consistency properties. In detail, we transform the middle feature $\mathbf{L}_{mid} \in \mathbb{R}^{U \times V \times H \times W \times C}$ to horizontal and vertical EPI features ($\mathbf{L}_{EPI-H} \in \mathbb{R}^{UH \times V \times W \times C}$, and $\mathbf{L}_{EPI-V} \in \mathbb{R}^{VW \times U \times H \times C}$), and apply DATLs to extract spatial-angular correlations sequentially

$$\begin{aligned} \mathbf{L}'_{EPI-H} &= \mathcal{F}_{DATL}^2(\mathcal{F}_{DATL}^1(\mathbf{L}_{EPI-H})), \\ \mathbf{L}'_{EPI-V} &= \mathcal{F}_{DATL}^2(\mathcal{F}_{DATL}^1(\mathbf{L}_{EPI-V})). \end{aligned} \quad (5)$$

So far, we have finished extracting domain-specific features of 4D LFs. However, the enhanced feature $\mathbf{L}'$ presents more structure information while the spatial context details are weakened as demonstrated in the ablation study of DAFEB, which may affect the final reconstruction quality. Therefore, to mitigate this, we lastly produce domain-agnostic features through the following operations. We first extract the discrepancies between $\mathbf{L}_{mid}$ and $\mathbf{L}'$, $\mathbf{L}_{dis} = \mathbf{L}_{mid} - \mathbf{L}'$, where $\mathbf{L}_{dis}$ mainly contain the overall structure information of LFs, then $\mathbf{L}_{dis}$ will be transferred to the former DATLs further re-enhance the spatial details of $\mathbf{L}_{dis}$ under robust angular consistency. Finally, the enhanced feature $\hat{\mathbf{L}}_{dis}$ will be added with the middle feature with rich spatial context information to obtain the final domain-agnostic output features with enhanced spatial details and better angular consistency.

$$\begin{aligned} \hat{\mathbf{L}}_{dis}^{MacPI} &= \mathcal{F}_{DATL}^2(\mathcal{F}_{DATL}^1(\mathbf{L}_{dis}^{MacPI})), \\ \hat{\mathbf{L}}_{dis}^{SAI} &= \mathcal{F}_{DATL}^2(\mathcal{F}_{DATL}^1(\mathbf{L}_{dis}^{SAI})), \\ \mathbf{L}_{out} &= \hat{\mathbf{L}}_{dis} + \mathbf{L}_{mid}. \end{aligned} \quad (6)$$

Remark. Before each subspace features are processed by DATLs, we utilize pixel attention (Zhao et al. 2020) to inject conditional positional encoding (kernel size of depth-wise convolution is set to 3×3) for better performance.

Experiment

Datasets and Implementation Details

Following prior works (Wang et al. 2023), we select the BsicLFSR benchmark which contains five LF datasets (i.e., EPFL (Rerábek and Ebrahimi 2016), HCIold (Wanner, Meister, and Goldlücke 2013), HCInew (Honauer et al. 2016), INRIA (Pendur, Jiang, and Guillemot 2018), STFgantry (Vaish and Adams 2008)) for training and testing. Specifically, we select the central 5×5 SAIs and crop them into $32 \times 32/64 \times 64$ patches for $\times 2/\times 4$ SR. Then, we perform bicubic downsampling to generate low-resolution LF images. Similar to (Wang et al. 2023), data augmentation is conducted by random rotation and flipping. In the training phase, we select Adam as the optimizer and L_1 as the loss function. The batch size is 8 and the initial learning rate is set to 4×10^{-4} and will be halved every 15 epochs in total 70 epochs. For testing, the LF image is also cropped into 32×32 patches for saving memory. All experiments are conducted on a single Nvidia 4090D GPU.

Method	Scale	#Param.(M)	FLOPs(G)	EPFL	HCInew	HCIdold	INRIA	STFgantry	Average
Bicubic	$\times 2$	-	-	29.50/9350	31.69/9335	37.46/9776	31.10/9563	30.82/9473	31.11/9542
resLF	$\times 2$	7.98	37.06	32.75/9672	36.07/9715	42.61/9922	34.57/9784	36.89/9873	36.58/9793
LFSSR	$\times 2$	0.88	25.70	33.69/9748	36.86/9753	43.75/9939	35.27/9834	38.07/9902	37.73/9835
LF-InterNet	$\times 2$	5.04	47.46	34.14/9761	37.28/9769	44.45/9945	35.80/9846	38.72/9916	38.08/9847
LF-ATO	$\times 2$	1.22	597.66	34.27/9757	37.24/9767	44.20/9942	36.15/9842	39.64/9929	38.15/9843
MEG-Net	$\times 2$	1.69	48.40	34.34/9773	37.42/9777	44.08/9942	36.09/9849	38.77/9915	38.14/9851
LF-DFNet	$\times 2$	3.94	57.22	34.44/9766	37.44/9786	44.23/9943	36.36/9841	39.61/9935	38.41/9854
IINet	$\times 2$	4.84	56.16	34.68/9773	37.74/9790	44.84/9948	36.57/9853	39.86/9936	38.74/9857
LF-SAV	$\times 2$	1.22	34.65	34.62/9772	37.43/9776	44.22/9942	36.36/9849	38.69/9914	38.26/9851
DistgSSR	$\times 2$	3.53	64.11	34.81/9787	37.96/9796	44.94/9949	36.59/9859	40.40/9942	38.94/9866
HLFSR	$\times 2$	13.72	167.40	35.31/9800	38.32/9807	44.98/9950	37.06/9867	40.85/9947	39.30/9874
DPT	$\times 2$	3.73	65.34	34.48/9758	37.35/9771	44.31/9943	36.40/9843	39.52/9926	38.40/9848
LFT	$\times 2$	1.11	56.16	34.80/9781	37.84/9791	44.52/9945	36.59/9855	40.51/9941	38.85/9863
EPIT	$\times 2$	1.42	69.71	34.83/9775	38.23/9810	45.08/9949	36.67/9853	42.17/9957	39.40/9877
LF-DET	$\times 2$	1.59	48.50	35.26/9797	38.31/9807	44.99/9950	36.95/9864	41.76/9955	39.45/9875
LF-DANet (Ours)	$\times 2$	1.30	48.08	36.30/9839	38.49/9814	45.32/9952	37.71/9883	41.54/9954	39.90/9888
Bicubic	$\times 4$	-	-	25.14/8311	27.61/8507	32.42/9335	26.82/8860	25.93/8431	27.58/8701
resLF	$\times 4$	8.64	39.70	28.27/9035	30.73/9107	36.71/9682	30.34/9412	30.19/9372	31.25/9322
LFSSR	$\times 4$	1.77	128.44	28.27/9118	30.72/9145	36.70/9696	30.31/9467	30.15/9426	31.23/9370
LF-InterNet	$\times 4$	5.48	50.10	28.67/9162	30.98/9161	37.11/9716	30.64/9491	30.53/9409	31.58/9388
LF-ATO	$\times 4$	1.36	686.99	28.52/9115	30.88/9135	37.00/9699	30.71/9484	30.61/9430	31.54/9373
MEG-Net	$\times 4$	1.77	102.20	28.74/9160	31.10/9177	37.28/9716	30.66/9490	30.77/9453	31.71/9399
LF-DFNet	$\times 4$	3.99	57.31	28.77/9165	31.23/9196	37.32/9718	30.83/9503	31.15/9494	31.86/9415
IINet	$\times 4$	4.88	57.42	29.11/9188	31.36/9208	37.62/9734	31.08/9515	31.21/9502	32.08/9429
LF-SAV	$\times 4$	1.54	115.80	29.37/9223	31.45/9217	37.50/9721	31.27/9531	31.36/9505	32.19/9439
DistgSSR	$\times 4$	3.58	65.41	28.99/9195	31.38/9217	37.56/9732	30.99/9519	31.65/9535	32.11/9440
HLFSR	$\times 4$	13.87	182.52	29.20/9222	31.57/9238	37.78/9742	31.24/9534	31.64/9537	32.29/9455
DPT	$\times 4$	3.78	66.55	28.93/9170	31.19/9188	37.39/9721	30.96/9503	31.14/9488	31.92/9414
LFT	$\times 4$	1.16	57.60	29.25/9210	31.46/9218	37.63/9735	31.20/9524	31.86/9548	32.28/9447
EPIT	$\times 4$	1.47	74.96	29.34/9197	31.51/9231	37.68/9737	31.37/9526	32.18/9571	32.40/9452
LF-DET	$\times 4$	1.69	51.20	<u>29.47/9230</u>	<u>31.56/9235</u>	<u>37.84/9744</u>	<u>31.39/9534</u>	<u>32.14/9573</u>	<u>32.48/9463</u>
LF-DANet (Ours)	$\times 4$	1.35	49.35	30.02/9280	31.84/9268	38.09/9755	31.98/9565	32.71/9609	32.93/9495

Table 1: **Quantitative comparison (PSNR / SSIM) of different methods for $\times 2$ and $\times 4$ SR.** The best and the second best results are **highlighted** and underlined, respectively. The parameters (#Param.) and FLOPs are calculated on an input light field with the size $5 \times 5 \times 32 \times 32$.

Comparison with the State-of-the-arts

We compare our method with fourteen state-of-the-art methods, including ten CNN-based LFSR methods resLF (Zhang, Lin, and Sheng 2019), LFSSR (Yeung et al. 2019), LF-InterNet (Wang et al. 2020), LF-ATO (Jin et al. 2020), MEG-Net (Zhang, Chang, and Lin 2021), LF-DFNet (Wang et al. 2021), IINet (Liu et al. 2023), LF-SAV (Cheng, Liu, and Xiong 2022), DistgSSR (Wang et al. 2023), and HLFSR (Duong et al. 2023), and four Transformer-based methods DPT (Wang et al. 2022a), LFT (Liang et al. 2021), EPIT (Liang et al. 2023), LF-DET (Cong et al. 2024). The most commonly used PSNR and SSIM are chosen to evaluate the performance. In detail, the metric scores of one scene are obtained by averaging over all SAIs, and that of one dataset by averaging over all scenes. We select bicubic upsampling as the baseline.

Quantitative Comparison. The quantitative results achieved by different methods are reported in Table 1. Our method almost obtains the best metrics on both $\times 2$ and $\times 4$ SR across five datasets. Specifically, our method outperforms the second best method LF-DET (Cong et al. 2024) with a significant improvement of 0.45/0.45 dB in

average PSNR and 0.0013/0.0032 in SSIM for the $\times 2$ and $\times 4$ tasks, respectively. This is mainly because our proposed DATL is more effective in extracting fine-grained features. Notably, while our method outperforms LF-DET and EPIT on the STFgantry dataset for $\times 4$ SR, it falls behind them on the $\times 2$ SR task. This discrepancy is likely due to our decision to use a small stripe width in DATL to enhance local feature awareness. Although this improves the model’s ability to capture fine details, it may limit its capacity to capture long-range pixel correspondences, which are more critical in scenarios with larger disparities. In the $\times 2$ SR task, where the input LF images exhibit larger disparities and less severe spatial information loss compared to the $\times 4$ SR task, LF-DET and EPIT are good enough at recovering local spatial details inside a SAI, whereas our model finds it more challenging to capture long-term pixel correspondences between SAIs. Additionally, to test the generalization abilities of different models, we use pre-trained models to test on the UrbanLF (Sheng et al. 2022) dataset. For detail results, see Appendix.

Visual Comparison. Visual comparisons among different methods are presented in Fig. 4, we select two chal-

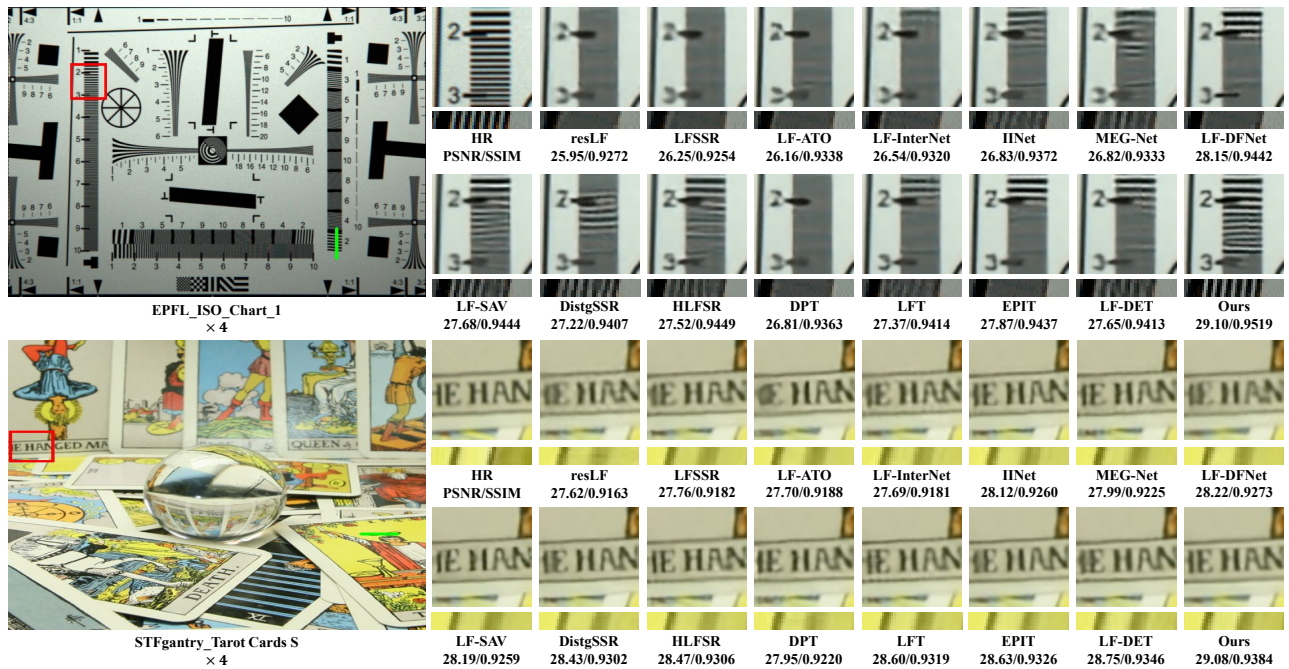


Figure 4: **Visual comparisons of different methods for $\times 4$ SR.** We present the close-up in the red box region and the EPI extracted from the green line. Best viewed zoom-in.

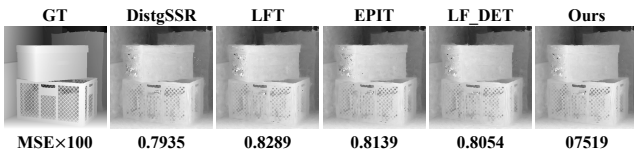


Figure 5: Quantitative and qualitative (MSE) comparisons of depth estimation results achieved by SPO (Zhang et al. 2016) using different $\times 4$ SR results.

lensing scenes ISO_Chart_1 from EPFL dataset and Tarot_Cards_S from STFgantry dataset on $\times 4$ SR task. We can observe that in scene ISO_Chart_1, only our method can well reconstruct the scale lines between figure ‘2’ and ‘3’. In scene Tarot_Cards_S, our method recovers more clear characters compared to other methods, demonstrating our method’s superiority and effectiveness in recovering detailed textures. More visual results can be found in the Appendix. In addition, the EPIs extracted from our method obtain better fidelity, and the depth estimation result using the SR LF images by our method achieves lower MSE as shown in Fig. 5, which indicates our method’s high angular consistency, demonstrating the effectiveness of our proposed DAFEB.

Computational Efficiency. We also evaluate the computation efficiency of our method by comparing the parameters, FLOPs, and inference time to other methods in $\times 4$ SR task. The parameters and FLOPs are listed in Table. 1, thanks to CW-SA, our method enjoys a moderate model size with only 1.35M parameters, which is only higher than

Method	DistgSSR	DPT	LFT	EPIT	LF-DET	Ours
Time (s)	0.28	0.47	0.54	0.68	0.62	0.37

Table 2: The inference time of different LFSR methods. Tested on Nvidia 4090D GPU with an $5 \times 5 \times 32 \times 32$ input LF for $\times 4$ SR.

LFT (1.16M). In terms of FLOPs, our method have the lowest FLOPs among all Transformer-based methods, and also lower than most CNN-based methods except resLF. The inference time of different methods are shown in Table. 2, we choose DistgSSR as the baseline and existing Transformer-based methods for comparison. As we can see, our method achieves faster inference time than existing Transformer-based methods and is close to DistgSSR, exhibiting our method’s efficiency.

Ablation Study

Detail-Awareness Transformer Layer. To investigate the effectiveness of the Detail-Awareness Transformer Layer, we gradually add CW-SA and CG-FFN from the vanilla Transformer and test on two real-world datasets INRIA and STFgantry, which the former is taken outdoors and the latter contains close-up objects that have larger disparity. As shown in Table. 3, removing CG-FFN from the full model leads to notable performance drops (0.12dB↓ on INRIA and 0.09dB↓ on STFgantry), and adding it to the vanilla Transformer can bring a marginal improvement, which showcases the efficacy of CG-FFN. On the other side, removing CW-SA from the full model causes a dramatic performance de-

Setting		#Param.(M)	PSNR	
CW-SA	CG-FFN		INRIA	STFgantry
		1.76	31.66	32.52
	✓	1.79	31.71	32.56
✓		1.32	31.86	32.62
✓	✓	1.35	31.98	32.71

Table 3: Ablation study on DATL for $\times 4$ SR.

Model Variants	#Param.(M)	FLOPs (G)	Average
<i>w/o re-enhance</i>	1.35	33.16	32.82
<i>w/ fusion</i>	1.36	33.90	32.67
<i>w/o weight-sharing</i>	1.99	49.35	32.92
<i>w/ sequential</i>	1.32	48.79	32.84
<i>w/ local-residual</i>	1.32	48.79	32.73
<i>w/ global-residual</i>	1.32	48.79	32.90
<i>full model</i>	1.35	49.35	32.93

Table 4: Ablation study on the designs of DAFEB and the residual channel attention learning in DAFEG for $\times 4$ SR.

crease (0.27dB $\downarrow$ on INRIA and 0.15dB $\downarrow$ on STFgantry), and adding CW-SA to the vanilla Transformer results in 0.20dB and 0.10dB gains on INRIA and STFgantry, respectively, demonstrating that CW-SA is better at exploring fine-grained spatial context information.

Domain-Agnostic Feature Extraction Block. We introduce three model variants to validate the effectiveness of DAFEB as shown in Table. 4. For model *w/o re-enhance*, we only preserve the first (spatial details enhancement) and the second steps (angular consistency enhancement) in DAFEB with a residual connection. As shown in Table. 4, the performance decreases by 0.11dB on average, and from Fig. 6 we can observe that the features exhibit coarse spatial details since they end up with an EPI feature extraction. Then, we introduce model *w/ fusion* which parallelly extracts spatial features and EPI features and then fuse them by concatenation and convolution. The performance of this model decreases significantly by 0.25dB, and the fused features do not successfully incorporate the angular consistency properties as illustrated in Fig. 6. In addition, we investigate our choice of utilizing the previous DATLs by conducting new corresponding DATLs as model *w/o weight-sharing*. The result shows that additional layers (or parameters) bring no improvement to the overall performance, demonstrating the effectiveness of our design.

Residual Channel Attention Learning. We further study the effect of Residual Channel Attention Learning. Three model variants are developed and the results are reported in Table. 4. Specifically, model *w/ sequential* removes the Residual Channel Attention Learning and simply cascades $N_1 N_2$ DAFEBs, which leads to a 0.09dB drop in average PSNR. If we replace the Residual Channel Attention Learning with a local residual connection in the DAFEG, it even

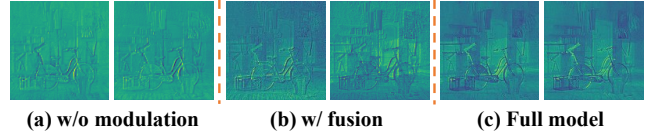


Figure 6: Comparisons of the feature maps extracted from the second DAFEB in the first DAFEG of the three models. The left one denotes the left-up corner SAI, and the right one denotes the right-down corner SAI.

N_1	N_2	#Param.(M)	FLOPs(G)	Average
1	2	0.76	27.05	32.67
1	3	1.07	38.48	32.74
2	2	1.35	49.35	32.93
2	3	1.94	71.92	33.09
3	2	1.93	71.64	33.04

Table 5: Investigation on different configurations of N_1 and N_2 . The results are tested on $\times 4$ SR.

performs poorer than model (c). Lastly, we add a global residual connection based on model *w/ sequential*, which can improve the performance but still slightly inferior to the full model, which demonstrates the effectiveness of our proposed Residual Channel Attention Learning. In addition, since Residual Channel Attention Learning is related to the organization of DAFEBs, we also evaluate the effect of different configurations of the number of DAFEBs (N_1) and DAFEGs (N_2). The results are presented in Table. 5. It can be observed that increasing DAFEGs can improve the performance consistently. To strike a balance between performance and efficiency, we ultimately set $N_1 = N_2 = 2$.

Conclusion and Limitations

Conclusion. In this paper, we propose a novel network, LF-DANet for LFSR. Specifically, we first develop DATL to facilitate the capture of local fine-grained details in LF features. Based on DATLs, we further propose DAFEB to learn domain-agnostic spatial-angular features from LFs, ensuring that the extracted features effectively preserve both spatial context information and robust angular consistency. We conduct extensive experiments to validate the effectiveness of our method, demonstrating that LF-DANet achieves superior performance on LF benchmarks, with the improved visual quality and higher quantitative metrics.

Limitations and Future Work. As our LF-DANet utilizes the proposed DATL as the general feature extractor on each LF's domains for fine-grained feature extraction as well as efficiency, it may encounter obstacles to capturing long-term correspondences on EPIs, which is more crucial in scenarios with large disparities. In the future, we will investigate the impact of the stripe width of DATLs employed in each domain feature extraction. Additionally, we will explore whether our approach to producing domain-agnostic spatial-angular features can be effectively applied to CNN-based networks.

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